YUBO LI

Email: yubol@andrew.cmu.edu | Mobile: (858) 666-7737 | Website: https://yubol-bobo.github.io/

EDUCATION

Carnegie Mellon University - *Ph.D. in Information Systems* 2022 – Present

Carnegie Mellon University - *M.S. in Information Systems & Management* 2020 – 2021

University of California, San Diego - *B.S. in Applied Mathematics | Minor in Business* 2015 – 2019

Leadership: Founder of the UCSD Triton Actuarial Society | GPA:3.8/4.0

PUBLICATIONS

- Confidence as Control: A Survey of Confidence Utilization in Large Language Models. TMLR
- Consistency of Large Reasoning Models Under Multi-Turn Attacks. EMNLP 2026
- Same Question, Different Source, Different Answer: Auditing Source-Dependence in Medical Multi-Source RAG. EMNLP 2026
- The Chain Holds, the Answer Folds: Trace-Answer Dissociation in Reasoning Models Under Adversarial Pressure. EMNLP 2026
- LLMs Know the Constraint But Do Not Use It: Activation Bottlenecks in Pragmatic Constraint Reasoning. EMNLP 2026
- The Model Says Walk: How Surface Heuristics Override Implicit Constraints in LLM Reasoning. COLM 2026
- Skills-Bench : Benchmarking the Efficacy of Agent Skills Across Diverse Tasks, NeurIPS 2026
- *PANDO: Efficient Multimodal AI Agents via Online Skill Distillation.* NeurIPS 2026 (Under review)
- *Beyond Single-Turn: A Survey on LLM Multi-Turn Interactions*. TMLR 2026.
- Time-To-Inconsistency: A Survival Analysis of Large Language Model Robustness to Adversarial Attacks. ICLR 2026.
- *Firm or Fickle? Evaluating LLM Consistency Capability in Multi-Turn Interactions*. ACL 2025.
- *No Black Box Anymore: Demystifying Clinical Predictive Modeling with Temporal-Feature Cross Attention Mechanism*. American Medical Informatics Association 2025 Annual Symposium (3rd Place in Paper Competition).
- *Enhancing End Stage Renal Disease Outcome Prediction: A Multi-Sourced Data-Driven Approach*. Journal of the American Medical Informatics Association, 2025.
- Co-Alignment: Rethinking Alignment as Bidirectional Human-AI Cognitive Adaptation. Agents4Science 2025.
- *Towards Interpretable End-Stage Renal Disease (ESRD) Prediction: Utilizing Administrative Claims Data with Explainable AI Techniques*. American Medical Informatics Association 2024 Annual Symposium.

FELLOWSHIPS

- 2025 Presidential Fellowship, Tata Consultancy Services (TCS)
- 2024 Fellowship in Generative AI in Healthcare, Center for Machine Learning and Health, Carnegie Mellon University
- 2023 Fellowship in Digital Health Innovation, Center for Machine Learning and Health, Carnegie Mellon University

RESEARCH EXPERIENCE

- **Efficient AI Agents | Agent Skills** **Carnegie Mellon University**

  Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2025.01 - Present
  - **AI Agents:** Built PANDO, a single-rollout online skill-distillation framework that equips multimodal web agents with a structured Skill Library via progress reflection, confidence-based skill demotion, hierarchical routing, visual compression, and cache-aware prompting; reached 58.3% success on the full 910-task VisualWebArena while using 58-61% fewer tokens than prior agents (SGV, WALT).
  - **Agent Skills Benchmarking:** Co-developed SkillsBench, a paired no-Skills versus curated-Skills benchmark spanning 87 tasks across 8 domains with deterministic verifiers, showing curated skills lift average pass rate from 33.9% to 50.5% (+16.6 pp) and that focused, three-module skills outperform exhaustive bundles; also surveyed dynamic, self-evolving agent skills.
  - **Efficient Inference:** Designed CONF-KV, a confidence-aware KV-cache manager with mixed FP16/INT8 storage that sets per-step cache budgets from model uncertainty, achieving 91.4% Needle-in-a-Haystack retrieval at 32K tokens (versus 53.8% for sliding windows) at 2.8x lower peak memory.
- **Q&A Systems for Trustworthy Organ Transplantation Guidance** **Carnegie Mellon University**

  Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2025.01 - Present
  - **Q&A Systems:** Developed Q&A systems for organ transplantation patient care by integrating retrieval-augmented generation (RAG) with adaptive frameworks that synthesize patient-specific medical history, clinical guidelines, and dynamic risk factors to deliver personalized guidance and support.
  - **RAG/Retrieval:** Enginnered a hybrid retrieval pipeline combining both sparse and dense retrievers (DPR, ANCE) to accurately extract and verify metical content from over 100 transplant-specific handbooks.
  - **LLM Reasoning:** Developed an incentivized reasoning mechanism that align LLM behaviors by rewarding multi-step chain-of-question sequences, enabling the system to reason freely and proactively request critical diagnostic information.
  - **Source Auditing:** Built a framework to audit source-dependence in multi-source medical RAG, releasing TransplantQA and a hierarchical retrieve-and-audit strategy (HERO-QA) scored by a validated five-label inter-source taxonomy; across

102 handbooks from 23 transplant centers, quantified that 20.8% of comparable answers diverge clinically while 96.2% of question-source pairs reveal coverage gaps.

- **LLM Alignment & Multi-Turn Consistency** **Carnegie Mellon University**
  Supervisor: Ramayya Krishnan, Dean of the Heinz College | Rema Padman, CMU Professor 2024.05 - Present
  - **Distributed Training:** Architected and optimized distributed training infrastructure for LLMs (Mistral, LLaMA, Deepseek) across 4 nodes with 16 NVIDIA GH200 GPUs, implementing tensor parallelism, pipeline parallelism, and ZeRO optimizer techniques to efficiently scale model training and reduce computational overhead.
  - **Fine-tuning:** Performed fine-tuning of state-of-the-art LLMs both locally (on-premises clusters) and via OpenAI's official GPT-4o API, successfully improving model consistency, confidence-aware reasoning, and achieving state-of-the-art performance.
  - **Evaluation:** Proposed a metric to quantitatively assess multi-turn response stability, prioritizing early interaction accuracy and swift recovery from initial errors. Released a comprehensive benchmark to test the model's ability to sustain coherent and contextually accurate responses during dynamic, multi-turn conversations.
  - **Faithfulness & Constraints:** Documented unfaithful capitulation, where a correct chain-of-thought persists while the emitted answer flips wrong under pressure, and introduced the Heuristic Override Benchmark showing salient surface cues override unstated feasibility constraints; released the supporting benchmarks, trajectories, and judge labels.

WORKING EXPERIENCE

Netflix – *ML Research Scientist Intern* 2026.06 – 2026.08
- **Personalized GenRec:** Developed personalized generative search models that jointly encode free-text queries and user interaction history to produce ranked semantic-ID recommendation lists, improving MRR@50 from 0.22 to 0.67.
- **Continued pretraining & post-training:** Built a two-stage adaptation pipeline—continued pretraining on domain corpora followed by multi-task post-training—to equip the LLM with query-only retrieval, semantic-ID retrieval, instruction following, and reasoning, using a loss-weighting scheme to balance the competing objectives.
- **Causal Evaluation & Robustness:** Proposed counterfactual evaluation protocols to disentangle whether models learn to use user history causally versus memorizing history patterns, and stress-tested robustness under noisy-history interventions.

Amazon – *Applied Scientist Intern* 2025.06 – 2025.08
- **Generative RecSys:** Developed User-LLM pipeline for personalized recommendation by pre-training custom user encoder on 172k customer interaction sequences, then fine-tuning through progressive alignment strategies (adapter-only, adapter + LLM, full fine-tuning) and pooling mechanisms.
- Achieved superior predictive performance (60.1% Hit Ratio@3, 67.5% NDCG@3) compared to direct prompting baselines while delivering **93.4% token reduction** and **64.8% cost savings**.
- **RAG/Search**: Architected production-ready RAG service infrastructure using FAISS HNSW indexing for 60k user embeddings to enhance recommendation contest at scale.
- **Distributed Training**: Engineered distributed training across multi-GPUs using FSDP with mixed-precision BF16.

SELECTED ACADEMIC ACTIVITIES

- **Invited Talk & Guest Lecture**
  - POMS 2026 | Invited Tutorial on LLM/AI Agent Measurement Science & Evaluations
  - GenAI in Healthcare | 94-706 Healthcare Information Systems| Spring 26
  - Coding Agents | 95-891 Introduction to AI | Spring 26
  - AI Agents | 95-891 Introduction to AI | Fall 25/Spring 26
  - Deep Learning for NLP | 95-891 Introduction to AI | Spring 25
- **Teaching Assistant**
  - 11-711 Advanced NLP with Prof. Graham Neubig.
  - 10-680/10-681 Mathematical & Computational Foundation for Machine Learning with Prof. Henry Chai.
  - 95-891 Introduction to Artificial Intelligence with Prof. David Steier.
  - 95-888 Data Focused Python with Prof. Michael Simko.
  - 95-865 Unstructured Data Analysis with Prof. George Chen.
  - 95-760 Decision Making under Uncertainty with Prof. David Choi.
- **Courses**
  - 10-714 Deep Learning Systems
  - 10-715 Advanced Introduction to Machine Learning
  - 10-716 Advanced Machine Learning: Theory and Methods
  - 10-703 Deep Reinforcement Learning
  - 10-725 Convex Optimization
  - 10-805 Machine Learning with Big Data
  - 15-513 Introduction to Computer System
  - 15-688 Practical Data Science
  - 11-868 LLM Systems
  - 11-711 Advanced NLP
  - 11-785 Introduction to Deep Learning
  - 11-777 Multi-Model Machine Learning
  - 36-707 Regression Analysis
  - 36-731/36-732 Causal Inference